

Plant Disease Synthesis

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Abstract— This study explores the use of image synthesis techniques to overcome data scarcity by generating artificial diseased leaf images using an enhanced CycleGAN architecture. Two variants were proposed: Model 1, which applies Sobel filtering to both input and output images, and Model 2, which integrates thresholding and Sobel filtering on the input and Sobel-only filtering on the output. The models were trained using segmented healthy images from the PlantVillage dataset and evaluated using Frechet Inception Distance (FID), Kernel Inception Distance (KID), and downstream classification accuracy. Experimental results show that Model 2 produced more realistic images and achieved better classification performance when used with MobileNetV2, reaching 77.50% accuracy in the Real + Synthetic dataset setting. Although ViT performed best with real-only data, the results demonstrate that synthetic data generated by Model 2 can effectively enhance lightweight classifiers. This work highlights the potential of GAN-based image generation to support plant disease recognition tasks and provides a foundation for future improvements in synthetic data augmentation.

Keywords—Plant Disease, Synthetic Images, CycleGAN, Unpaired Image to Image Transition

I. INTRODUCTION

Plants are critical to life on Earth, acting as primary producers in the food chain and contributing to food security, environmental balance, and economic sustainability. However, plant health is constantly threatened by diseases caused by fungi, bacteria, viruses, and environmental stressors. These diseases can lead to severe crop loss, with global agricultural production suffering estimated losses of 20–40% annually due to pests and diseases, amounting to over \$220 billion in economic impact.

Accurate and timely detection of plant diseases is vital to mitigate these losses. In recent years, deep learning-based classification systems, particularly Convolutional Neural Networks (CNNs), have shown promise in automating plant disease recognition. However, the performance of these models depends heavily on the availability of large and diverse datasets. Due to the sporadic nature of disease outbreaks and environmental dependencies, acquiring sufficient diseased leaf images remains a significant challenge. Moreover, existing datasets often lack species and symptom diversity, which limits the generalization capability of these models.

To address the scarcity of real-world diseased leaf images, image synthesis using generative models such as CycleGAN has emerged as a promising alternative. These models can generate artificial diseased versions of healthy leaves, enhancing dataset size and diversity without requiring costly field data collection. Despite their potential, challenges remain in generating realistic disease symptoms while preserving the structural integrity of leaves. Issues such as visual artifacts, mixed or overlapping disease features, and lack of symptom

diversity across species reduce the effectiveness of such models.

This study proposes a CycleGAN-based image synthesis framework aimed at generating high-fidelity diseased plant leaf images. The primary objectives are to (1) design a generative model capable of simulating realistic disease symptoms on healthy leaves, (2) evaluate the quality of the generated images using FID and KID metrics, and (3) test the synthesized images in classification tasks to measure their practical utility. Additional enhancements such as species-aware discrimination and shape-preserving loss functions are explored to improve the realism and usability of the generated images. The PlantVillage dataset is used as the foundation for training and evaluation. This research aims to contribute to more robust and scalable plant disease recognition systems by addressing the data scarcity problem through advanced image synthesis techniques.

II. RELATED WORK

The development of robust plant disease recognition systems depends heavily on the availability of high-quality datasets and advanced machine learning techniques. This section reviews key works in three areas relevant to this study: plant disease datasets, plant disease classification, and image synthesis using generative models.

A. Plant Disease Datasets

The **PlantVillage dataset**, proposed by Mohanty et al. [1], is among the most widely used resources in plant disease research. It contains over 54,000 images representing 38 classes of healthy and diseased leaves from 14 plant species, and is available in RGB, segmented, and grayscale formats. Its diversity and scale make it ideal for training deep learning models and performing various image synthesis experiments. Compared to datasets like the Kashmiri Apple Plant Disease Dataset [2] and the PlantDoc dataset [3], PlantVillage offers superior balance between sample size, disease variety, and image quality, despite being captured in controlled environments.

B. Plant Disease Classification Models

Recent research has demonstrated the effectiveness of deep learning models such as CNNs and Vision Transformers for plant disease classification. Models like VGG16 [4], MobileNet [5], and GSMo-CNN [6] have shown promising results on datasets including PlantVillage and PlantDoc. MobileNet, for example, achieved up to 96% accuracy while remaining lightweight and suitable for mobile deployment. GSMo-CNN extended these capabilities with multi-task learning, improving F1 scores to over 99% on the PlantVillage dataset. However, performance tends to degrade on

imbalanced or limited datasets, highlighting the need for augmentation techniques like image synthesis.

C. GAN-Based Image Synthesis

Generative Adversarial Networks (GANs) have emerged as powerful tools for image synthesis. The **CycleGAN** model introduced by Zhu et al. enables unpaired image-to-image translation, making it suitable for domains where paired healthy-diseased samples are unavailable. In the context of plant pathology, CycleGAN is advantageous because it maintains cycle consistency and identity mapping, ensuring that disease features are added without significantly altering the leaf structure. This makes it suitable for generating high-quality synthetic diseased images from healthy samples.

Several studies have explored GANs in plant disease synthesis. For instance, Dwarkani et al. [7] demonstrated the use of CycleGAN for leaf-to-leaf translation tasks. More recent models such as LeafyGAN [8] integrate Pix2Pix for segmentation and CycleGAN for disease translation, achieving enhanced realism and symptom accuracy. However, these models are often computationally expensive and may produce artifacts without careful tuning.

In this study, CycleGAN is adopted due to its ability to learn transformations from healthy to diseased leaves using unpaired images. Its flexibility and cycle consistency loss make it well-suited for generating synthetic training data, especially when paired datasets are scarce or unavailable.

III. PROPOSED METHOD

The core idea of using CycleGAN is because it is a model which specialise on unpaired image to image transition.

A. Overview of CycleGAN

CycleGAN, or Cycle-Consistent Generative Adversarial Network, is a type of GAN architecture designed for unpaired image-to-image translation. It consists of two generator networks GGG and FFF, and two discriminator networks DXD_XDX and DYD_YDY. The generators learn to translate images between two domains (e.g., healthy and diseased leaves), while the discriminators evaluate whether the generated images appear realistic within their respective domains.

To ensure consistency and reduce mode collapse, CycleGAN uses two key loss functions:

- **Adversarial Loss:** This helps the generator learn to create images that look so real, even the discriminator can't tell they're fake.
- **Cycle-Consistency Loss:** This makes sure that if an image is converted from one domain to another and then back again, it still looks like the original.

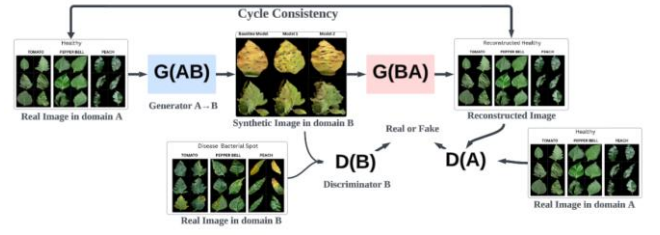


Figure 1: CycleGAN Architecture

This architecture is suitable for plant disease image synthesis because it allows realistic transformation of healthy leaves into diseased ones without requiring paired image datasets.

B. Enhancements to CycleGAN

To improve the disease pattern realism and structure preservation of the generated images, two enhanced models were proposed by modifying the input and/or output processing pipeline using Sobel edge filters and thresholding techniques.

Model 1: Dual Sobel CycleGAN

This model applies the Sobel edge operator to both the input healthy leaf image and the generated diseased output image. The goal is to enhance boundary and edge information during training, helping the network preserve the original leaf shape while applying disease patterns.

Model 2: Threshold-Sobel Input with Sobel Output

This model introduces a threshold + Sobel filter on the input healthy image and applies only the Sobel filter on the output. The thresholding highlights leaf regions with stronger contrast, and combining it with edge detection enhances the structural awareness of the generator. The output-side Sobel acts as a regularization technique, encouraging the preservation of clear edges and avoiding unnatural deformations.

C. Implementation Details

In this study, all images were resized to a standard input size of 256×256 pixels. For data selection, three plant species which is Peach, Bell Pepper, and Tomato were chosen from the PlantVillage segmented dataset. Only healthy and bacterial spot disease categories were used. The dataset was split into 80% for training and 20% for testing to ensure reliable performance evaluation. During preprocessing, the images were optionally converted to grayscale before applying thresholding and Sobel filtering, depending on the model variant.

The training of both enhanced CycleGAN models involved several loss functions: adversarial loss to encourage realistic image synthesis, discriminator loss to improve domain distinction, cycle consistency loss to preserve the original image content, identity loss to maintain stylistic consistency, and contour loss to enforce structure preservation, especially around leaf boundaries. The models were trained using the Adam optimizer with a learning rate of 0.0002, which decayed linearly over time. Each model was trained for 120 epochs with a batch size of 1.

The outputs were evaluated using quantitative metrics such as Frechet Inception Distance (FID) and Kernel Inception Distance (KID), as well as downstream classification performance, where the synthetic images were tested for their effectiveness in improving model accuracy when used in a plant disease recognition task.

IV. EXPERIMENTAL RESULTS

This section presents the performance evaluation of two enhanced CycleGAN models developed in FYP2: **Model 1** (Sobel applied to both input and output) and **Model 2** (threshold + Sobel on input, Sobel only on output). The models were evaluated based on their ability to generate realistic diseased images and their effectiveness in improving downstream plant disease classification performance.

A. Image Quality Evaluation

To assess the realism of the synthesized images, Frechet Inception Distance (FID) and Kernel Inception Distance (KID) were computed for both models.

Table 1: FID and KID scores for Model 1 and Model 2

Model	FID Score	KID Score
Baseline Model	33.32	0.67
Model 1	41.16	0.67
Model 2	37.95	0.68

Table 1 shows that the Baseline Model achieved the lowest FID score (33.32), followed by Model 2 (37.95) and Model 1 (41.16), indicating that Model 2 improved realism compared to Model 1. KID scores were similar across models (0.67–0.68), with Model 2 showing a slight increase, suggesting marginally reduced diversity. These results highlight a trade-off where Model 2 enhances perceptual realism but may require further tuning to improve feature-level variation.

B. Downstream Classification Accuracy

To evaluate the effectiveness of synthetic images in improving plant disease classification, two classifiers were used: Vision Transformer (ViT) and MobileNetV2. Each classifier was trained using three dataset configurations:

- All Real: Contains only real healthy and diseased leaf images from PlantVillage.
- Real + Synthetic: A mixed dataset that includes real healthy and diseased images along with synthetic diseased images generated by the models.
- All Synthetic: Real healthy images paired only with synthetic diseased images (no real diseased samples).

Table 2: Classification accuracy (%) using different dataset types and classifiers

Dataset Type	ViT Accuracy	MobileNetV2 Accuracy
All Real	98.08	72.25
Real + Synthetic (Model 1)	96.28	41.62
Real + Synthetic (Model 2)	95.09	77.50
All Synthetic (Model 1)	93.91	38.86
All Synthetic (Model 2)	95.54	61.25

Model 2 consistently outperformed Model 1 across both classifiers and all dataset types. The highest accuracy was obtained using **Model 2's synthetic images combined with real data**, demonstrating that high-quality synthetic augmentation can enhance model generalization and robustness.

C. Visual Analysis of Synthesized Images

To complement the quantitative evaluation of image quality metrics, we conducted a qualitative visual analysis of the synthesized diseased leaves across three plant species which is Peach, Pepper, and Tomato were generated by the baseline CycleGAN and two enhanced variants, Model 1 and Model 2. The models were assessed for realism, structural consistency, and alignment with known disease characteristics.

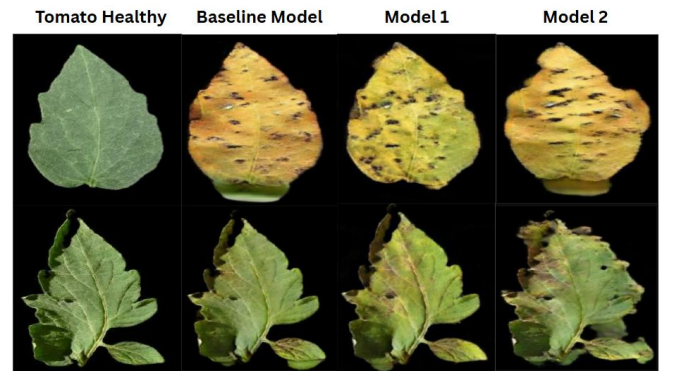


Figure 2: Comparison Tomato synthesis images between different models.

For tomato leaves, the baseline model produced slightly blurred and collapsed contours, but the internal lesion distribution was realistic. Lesions varied in density and occasionally mimicked necrosis or insect induced damage, giving an impression of natural pathogen spread. Model 1 demonstrated stronger edge and vein preservation while retaining a realistic distribution of necrotic spots. However, these were sometimes too dense, leading to unnatural perforation of the leaf surface. Model 2, while successful in adding tactile textures, often sacrificed leaf shape integrity. Lesions took on artificial forms such as thin elongated stripes and pit-like holes, deviating from the natural look of tomato leaf infections.

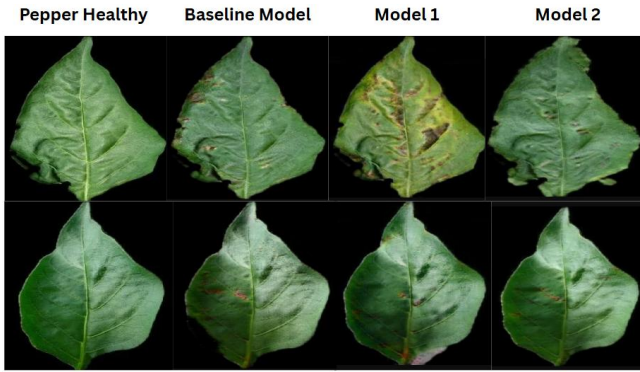


Figure 3: Comparison Pepper synthesis images between different models.

In the case of pepper leaves, the baseline CycleGAN introduced moderate edge deformation, especially along the margins, and produced spot patterns that appeared repetitive and synthetic. Model 1 improved shape preservation, with cleaner outlines and more vivid disease manifestations. However, these spots often resembled burn marks rather than biological symptoms, possibly due to the overly sharp edge enhancement. Model 2 introduced more variation in spot placement, particularly in the first row samples, which appeared biologically plausible. However, its performance was less consistent, like second row images featured faint and indistinct spots, reducing the overall clarity and diagnostic usefulness of the synthesized leaves.



Figure 4: Comparison Peach synthesis images between different models.

For peach leaves, the baseline CycleGAN model generated images with smooth, natural looking contours and a realistic green tint. However, it exhibited limitations in detail diversity, often producing overly uniform color patterns and “phantom” lobes, additional leaf structures do not present in the original images. Model 1 (Dual Sobel) enhanced plaque texture detail significantly, yielding sharp and crisply defined vein structures. Nonetheless, this came at the cost of overly etched edges and the introduction of an unnatural crimson hue in some samples, diverging from known peach disease appearances. Model 2 (Threshold + Sobel Input, Sobel

Output) offered a balance between the two, preserving the fluid contouring of the baseline while reducing the harshness of Model 1. It produced more believable yellow mottling patterns, though it lacked the fine textural details Model 1 excelled at and still occasionally introduced phantom leaflets.

V. CONCLUSION

This paper presented two enhanced CycleGAN models developed in FYP2 for the purpose of synthesizing realistic diseased plant leaf images. Model 1 applied Sobel filtering to both the input and output images, while Model 2 combined thresholding and Sobel filtering on the input with Sobel-only processing on the output. The goal was to generate structurally consistent synthetic images that could supplement training datasets for plant disease classification.

Quantitative evaluation using FID and KID scores showed that Model 2 produced more realistic images (FID = 37.95, KID = 0.68) compared to Model 1 (FID = 41.16, KID = 0.67). For downstream performance, two classifiers, Vision Transformer (ViT) and MobileNetV2—were trained using different dataset types. Results revealed that Model 2, when used in the Real + Synthetic setting, achieved the highest classification accuracy of 77.50% on MobileNetV2, outperforming Model 1. However, ViT performed best on the all real dataset with 98.08% accuracy, suggesting that real data still leads to optimal performance in some cases.

Overall, the results demonstrate that the use of synthetic images can enhance model robustness when combined with real data, particularly when using lightweight classifiers like MobileNetV2. Model 2's preprocessing strategy proved more effective, indicating that structure-preserving filtering techniques improve the quality of disease synthesis.

Future work may involve expanding the model to handle multi-disease conditions, testing across additional plant species, and exploring domain adaptation for real-world deployment.

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